

SHOU-JEN (LAWRANCE) CHEN

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EDUCATION

Electrical Engineering, National Taiwan University (NTU)

Taipei, Taiwan

Bachelor of Science in Engineering

Expected Graduation: Sep. 2020 - Jan. 2024

- Last-60-credit GPA: **4.27/4.3**
 - Dean's List: 2023 Fall (**Top 5%**)
- **Scholarship:** National Taiwan University Admissions Scholarship for Overseas Chinese Students
- **Relevant courses with A(+/-) grades:** *Machine Learning, Generative Artificial Intelligence, Data Structure, Introduction to Computer, Computer Networks, Wireless and Mobile Networking, Nano-Electronics, Electronic Circuits, Electronics(II), Integrated Circuit Design, Power Electronics Lab, Embedded System Lab, Electronics Lab, Electronic Circuits Lab Chemistry, Modern Physics etc... (* graduate-level)

RESEARCH EXPERIENCE

Robot Learning Laboratory (RLL), NTU

Taipei, Taiwan

Research Assistant & Student Researcher | Advisor: **Prof. Shao-Hua Sun**

Oct. 2023 – Present

- Introduced *LLM Discussion* framework that enhances the creativity of large language models via multi-agent discussion and role-playing, surpassing single-agent and existing multi-agent LLM frameworks in creativity tests by 20%.
- Designed an LLM evaluation focused on creativity metrics, achieving a 0.7 correlation with human evaluations on 1,400+ responses.
- Develop a continuous in-context learning (ICL) algorithm that optimizes decision-making for saving responses to Retrieval-Augmented Generation (RAG) and determining the appropriate learning method.
- Streamline updates by selecting between ICL, supervised fine-tuning, or reinforcement learning from human feedback (RLHF) to create an efficient, rolling learning process.

Speech Processing and Machine Learning Laboratory (SPMLLab), NTU

Taipei, Taiwan

Undergraduate Researcher | Advisor: **Prof. Hung-yi Lee**

Jul. 2023 – Present

- Devised and preprocessed a "SoundEffectDetection" dataset/task for Dynamic-SUPERB, aimed at building universal speech models capable of leveraging instruction tuning to perform multiple tasks in a zero-shot fashion. 
- Utilize Direct Preference Optimization (DPO) reinforcement learning as a method for Reinforcement Learning from Human Feedback (RLHF) to train a text-instruction-guided voice conversion model, aiming to enhance model performance.

Wireless and Mobile Networking Laboratory (WMNLab), NTU

Taipei, Taiwan

Undergraduate Researcher | Advisor: **Prof. Hung-Yu Wei**

Jan. 2023 – Jan. 2024

- Conducted mobile phone experiments on the MRT using user datagram protocol to evaluate network handover.
- Analyzed data and presented findings on signal strength and connectivity across a 25-km route to professors and graduate students.

PUBLICATIONS & CONFERENCE EXPERIENCE

- Li-Chun Lu*, **Shou-Jen Chen***, Tsung-Min Pai, Chan-Hung Yu, Hung-yi Lee, Shao-Hua Sun, "*LLM Discussion: Enhancing the Creativity of Large Language Models via Discussion Framework and Role-Play*", **COLM 2024. (Co-First Author)** 
- **Reviewer** for the **2024 IEEE Spoken Language Technology Workshop (SLT)**, Evaluated and provided feedback on research paper.
- Chien-yu Huang, Wei-Chih Chen, et al., **Shou-Jen Chen**, Shinji Watanabe, Hung-yi Lee, "*Dynamic-SUPERB Phase-2: An Open Benchmark Evolving through Collaborative Expansion for Comprehensive Evaluation of Spoken Language Models with 180+ Tasks*", **ICLR 2025. (On-going Submission)**

PROJECTS & INTERNSHIPS

Industry-University Collaboration, TrendForce

Taipei, Taiwan

Student Researcher

Oct. 2024 – Present

- Developing an automated system to generate technology news drafts, discussing topics specified by a given set of news articles.
- Exploring fine-tuning techniques and in-context example-based methods to align LLM outputs with the writing style of TrendForce news.

Software Engineer Team, Pegatron Corporation

Taipei, Taiwan

Software Engineer Intern

Jul. 2023 – Aug. 2023

- Collaborated in a team of three to develop “Visual Prompt Defect Detection” for an internal internship project competition, receiving **1st Place** among four total groups. Our project focuses on using in-context learning.
- Designed user interface using Gradio to present machine learning model’s capabilities for our project demo.
- Integrated APIs to automate the storage of large datasets in H5 format on the company’s server. This automation enabled easy loading and uploading of defect images for detection, improving data management efficiency.

Flyback Circuit Final Project

Individual Project

Feb. 2023 – Jun. 2023

- Designed and analyzed a complete flyback circuit, including layout analysis, transformer testing, and the use of filter and regulation capacitors.
- Conducted comprehensive testing, including DC and AC closed-loop tests, waveform analysis, and efficiency measurements, resolving technical challenges throughout the process.

Embedded System Lab Project 🤖

Group Leader

Aug. 2023 – Jan. 2023

- Collaborated with a teammate to develop a smart driving system, featuring real-time notifications and auto-parking.
- Integrated PiRacer Pro and STM32 board to boost system capabilities with analytics and machine learning.
- Presented and demonstrated our project to the professor and 25+ undergraduate students.

Software Engineer and Webpage Developer, IADIY Technology

Taipei, Taiwan

Intern

Jul. 2020 – Aug. 2020 / Jul. 2021 – Aug. 2021

- Designed software enabled auto-driving and basic sensor mapping using Robot Operating System on Linux.
- Assembled robotic devices, including constructing the hardware framework and configuring wires, running on Raspberry Pi.
- Initiated a Kickstarter Campaign in a team of 5 for the product, Genix Lite and designed the company website using HTML and CSS.

TEACHING EXPERIENCE

Generative Artificial Intelligence Course, NTU (Technical Report 📄 / Course Webpage 🌐)

Taipei, Taiwan

Teaching Assistant (TA)

Feb. 2024 – Jun. 2024

- Collaborated with a team of 30 TAs to manage and respond to student emails for a class of over 1,000 students and 1,000 auditors.
- Designed and graded a coursework related to calling API alongside two other TAs and provided 8 office hours.

English Written and Oral Communication for Professional Pursuit Course, NTU

Taipei, Taiwan

Teaching Assistant (TA)

Expected Date: Sept. 2024 – Jan. 2025

- Dedicate over 30 hours per week throughout the semester to support course instruction and student success.
- Manage and promptly respond to emails and inquiries for a class of over 70 students, ensuring clear and timely communication.
- Post weekly reminders and important updates to keep students informed and organized.

ACTIVITIES

Oversea Chinese Student Association, NTU

Taipei, Taiwan

Executive and Treasurer

Aug. 2022 – Jun. 2023

- Hosted over 6 successful events, engaging 30+ overseas students and fostering community connections.
- Managed committee finances efficiently, ensuring financial stability throughout the school year.
- Coordinated a prom with 7 executive members, attracting 60+ attendees and raising over \$80,000 while adhering to strict COVID-19 safety protocols.

Information Technology Department, NTU Electrical Engineering

Taipei, Taiwan

Executive and Treasurer

Aug. 2022 – Present

- Assisted in game and website design during events, focusing on frontend development with HTML and CSS.
- Participated in classes to develop programming skills, collaborate on IT projects, and strengthen technical expertise.

GMBA & FET Business Proposal Competition 2022 3rd Place, NTU MBA

Taipei, Taiwan

Group Member

May. 2022

- Collaborated with two teammates to create a business proposal focused on expanding the cloud service industry into Southeast Asia.
- Won 3rd place in the GMBA & FET Business Proposal Competition 2022 by presenting the proposal to a panel of judges.

Basketball Team, NTU Electrical Engineering

Taipei, Taiwan

Player

May. 2022

- Participated in athletic events, demonstrating teamwork and commitment to personal fitness and sportsmanship.
- Collaborated with teammates on and off the court to build strong communication and leadership skills.

LANGUAGE PROFICIENCY

- **Language:**
 - **English:** Mastery
 - **Chinese:** Native
 - **Thai:** Native

SKILLS

- **Programming Language:** Python, C++, Golang, HTML/CSS, Matlab
- **Miscellaneous:** LaTeX, Git, Gradio, Socket, Verilog, MongoDB, BLE, PSpice